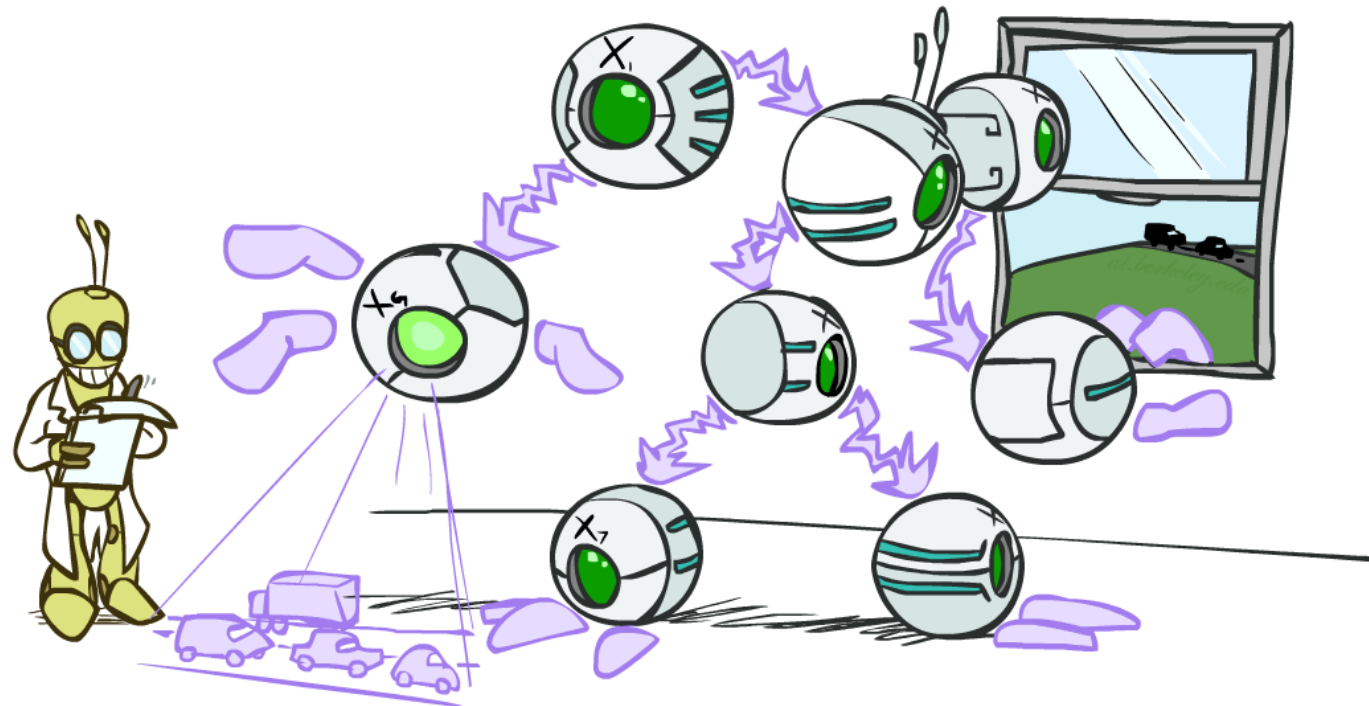


CS 6300: Artificial Intelligence

Bayes' Nets: Inference and Sampling



Instructor: Daniel Brown --- University of Utah

[Based on slides created by Dan Klein and Pieter Abbeel <http://ai.berkeley.edu>.]

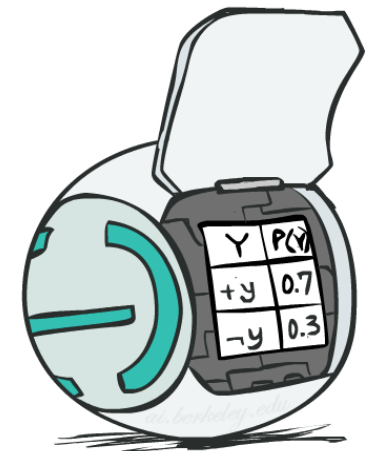
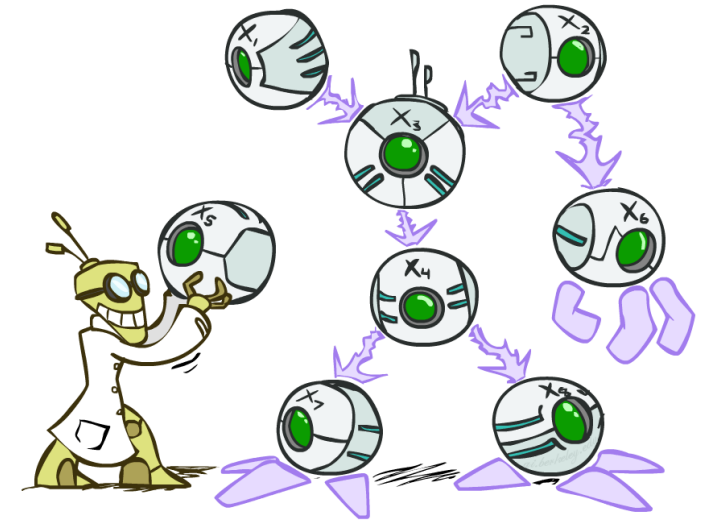
Bayes' Net Representation

- A directed, acyclic graph, one node per random variable
- A conditional probability table (CPT) for each node
 - A collection of distributions over X , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

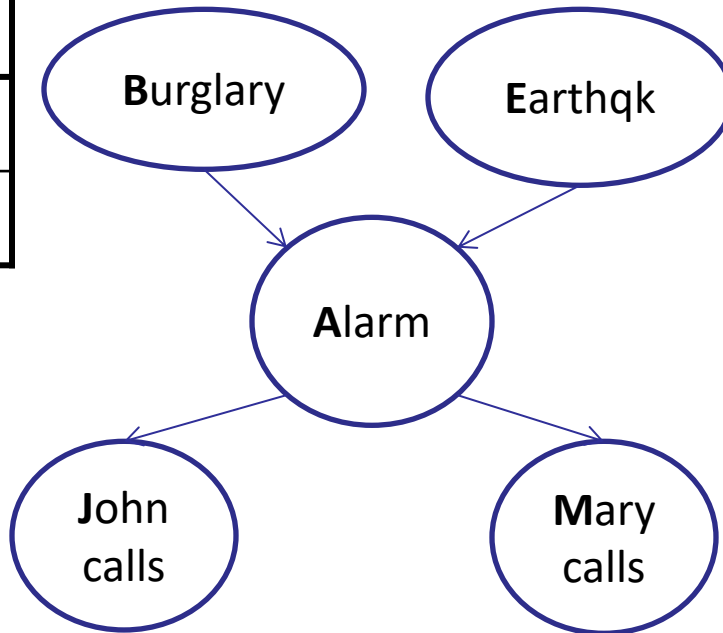
- Bayes' nets implicitly encode joint distributions
 - As a product of local conditional distributions
 - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

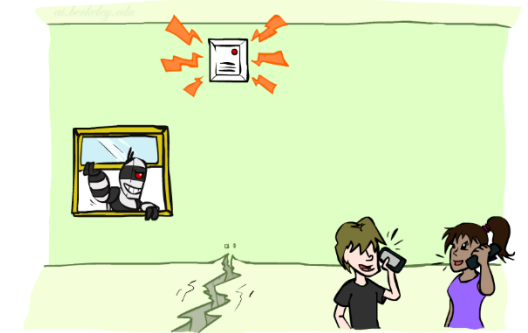


Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998



A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

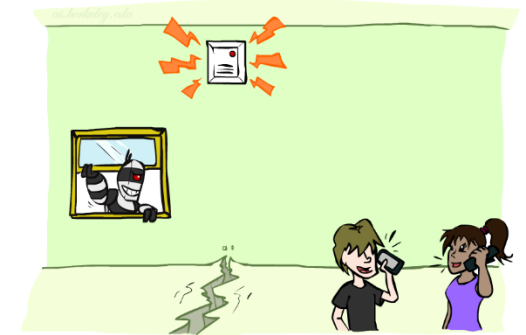
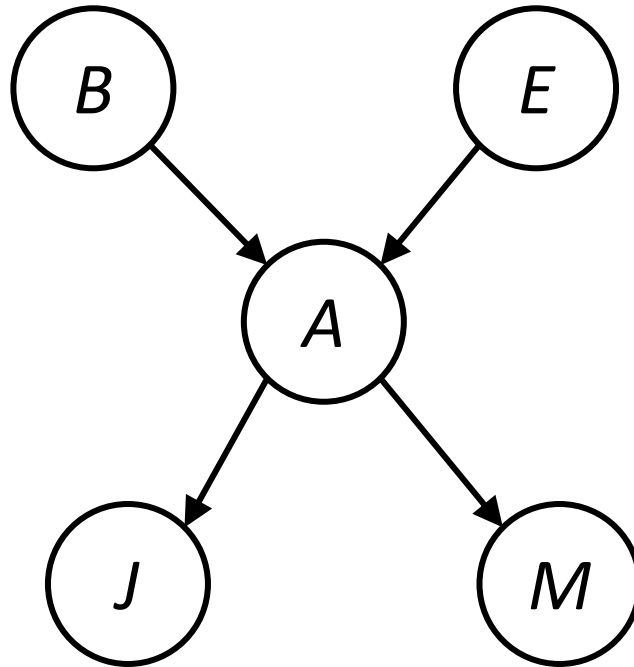
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+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, -e, +a, -j, +m) =$$

$$P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) =$$

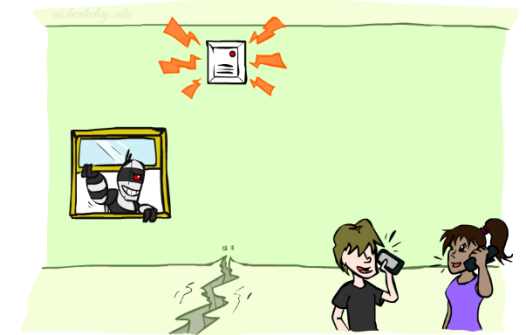
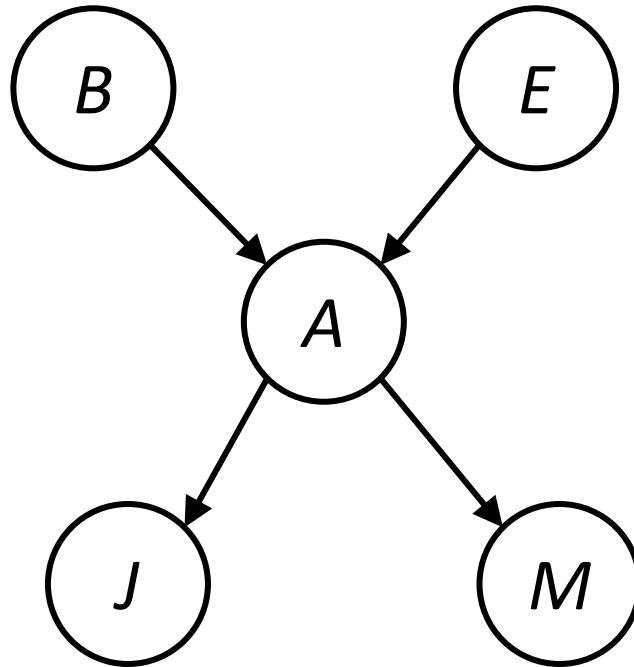
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+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$\begin{aligned}
 P(+b, -e, +a, -j, +m) &= \\
 P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) &= \\
 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7 &=
 \end{aligned}$$

Bayes' Nets



Representation



Conditional Independences



Probabilistic Inference

- Enumeration (exact, exponential complexity)
- Inference is NP-complete
- Sampling (approximate)

Inference

- Inference: calculating some useful quantity from a joint probability distribution

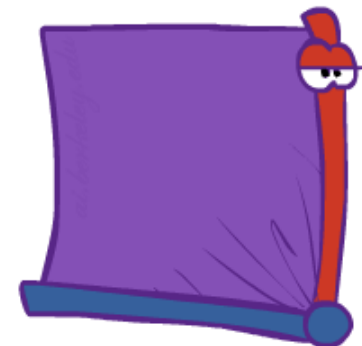
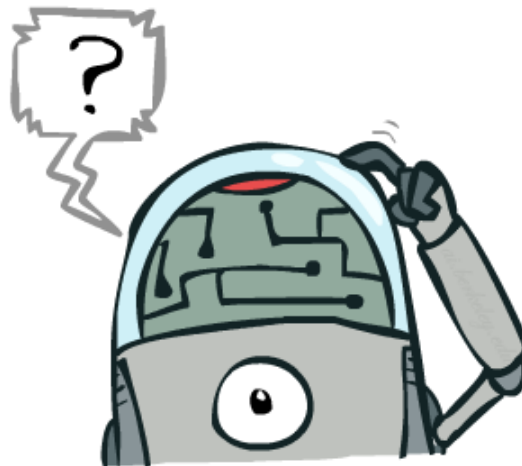
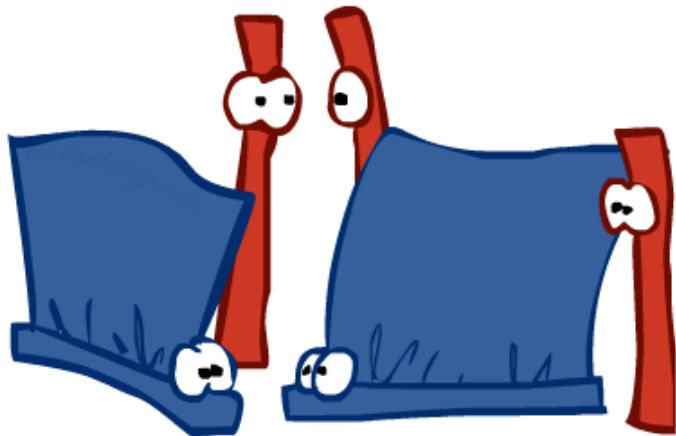
- Examples:

- Posterior probability

$$P(Q|E_1 = e_1, \dots, E_k = e_k)$$

- Most likely explanation:

$$\operatorname{argmax}_q P(Q = q|E_1 = e_1 \dots)$$



Inference by Enumeration

- General case:

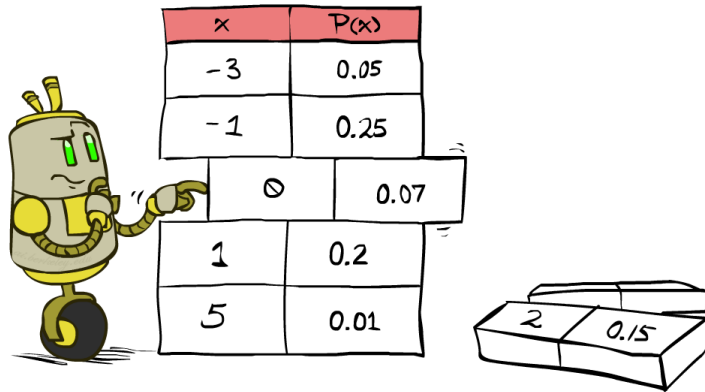
- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query* variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $$\left. \begin{array}{l} E_1 \dots E_k = e_1 \dots e_k \\ Q \\ H_1 \dots H_r \end{array} \right\} \begin{array}{l} X_1, X_2, \dots X_n \\ \text{All variables} \end{array}$$

- We want:

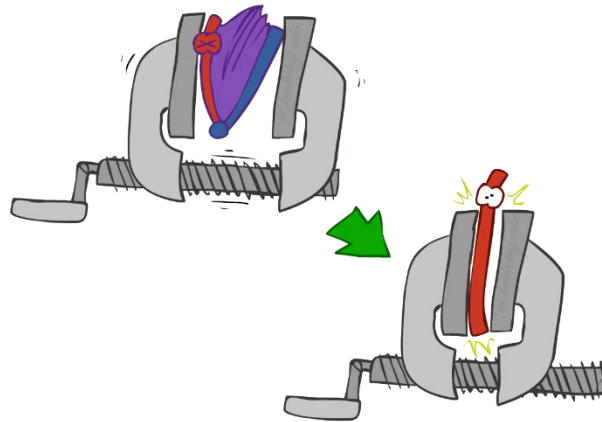
* Works fine with multiple query variables, too

$$P(Q|e_1 \dots e_k)$$

- Step 1: Select the entries consistent with the evidence



- Step 2: Sum out H to get joint of Query and evidence



$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} P(Q, \underbrace{h_1 \dots h_r}_{X_1, X_2, \dots X_n}, e_1 \dots e_k)$$

- Step 3: Normalize

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

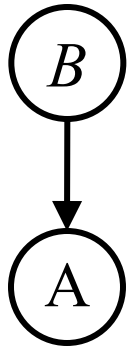
$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

Example Inference: $P(B|+a)$

Start / Select

$P(B)$

B	P
+b	0.1
$\neg b$	0.9



$P(A|B) \rightarrow P(a|B)$

B	A	P
+b	+a	0.8
b	$\neg a$	0.2
$\neg b$	+a	0.1
$\neg b$	$\neg a$	0.9

Normalize

$P(+a|B)P(B)$

A	B	P
+a	+b	0.08
+a	$\neg b$	0.09

$P(B|a)$

A	B	P
+a	+b	8/17
+a	$\neg b$	9/17

Inference by Enumeration in Bayes' Net

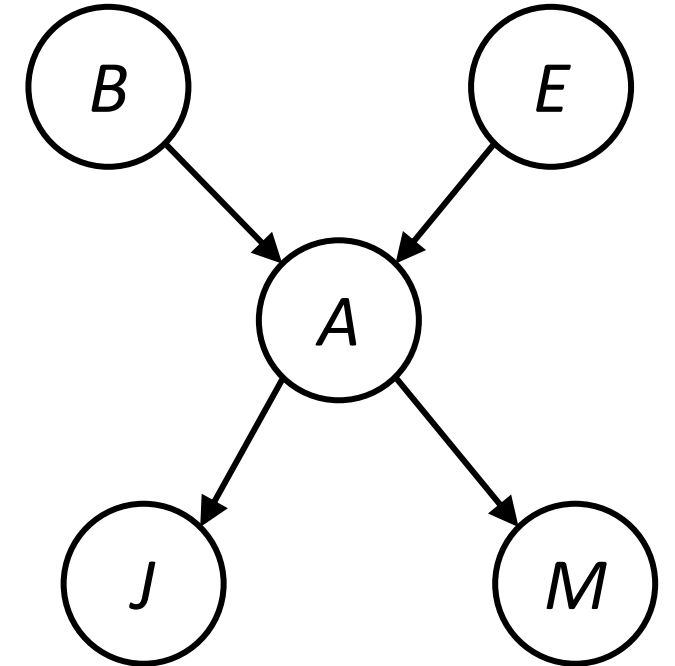
- Given unlimited time, inference in BNs is easy
- Reminder of inference by enumeration by example:

$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

$$= \sum_{e,a} P(B, e, a, +j, +m)$$

$$= \sum_{e,a} P(B)P(e)P(a|B, e)P(+j|a)P(+m|a)$$

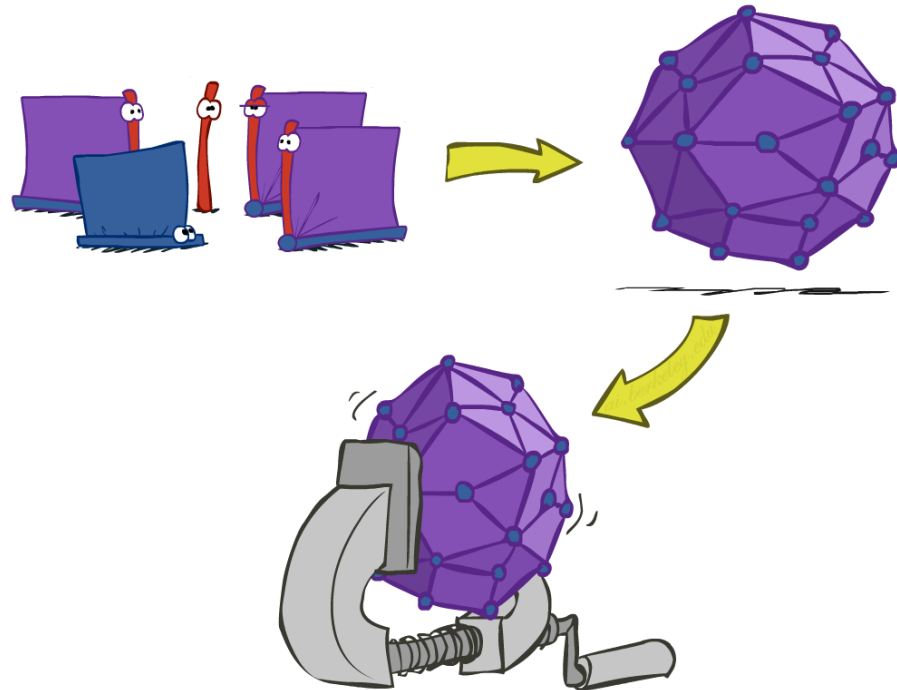
$$= P(B)P(+e)P(+a|B, +e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B, +e)P(+j|-a)P(+m|-a) \\ + P(B)P(-e)P(+a|B, -e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B, -e)P(+j|-a)P(+m|-a)$$



Inference by Enumeration vs. Variable Elimination

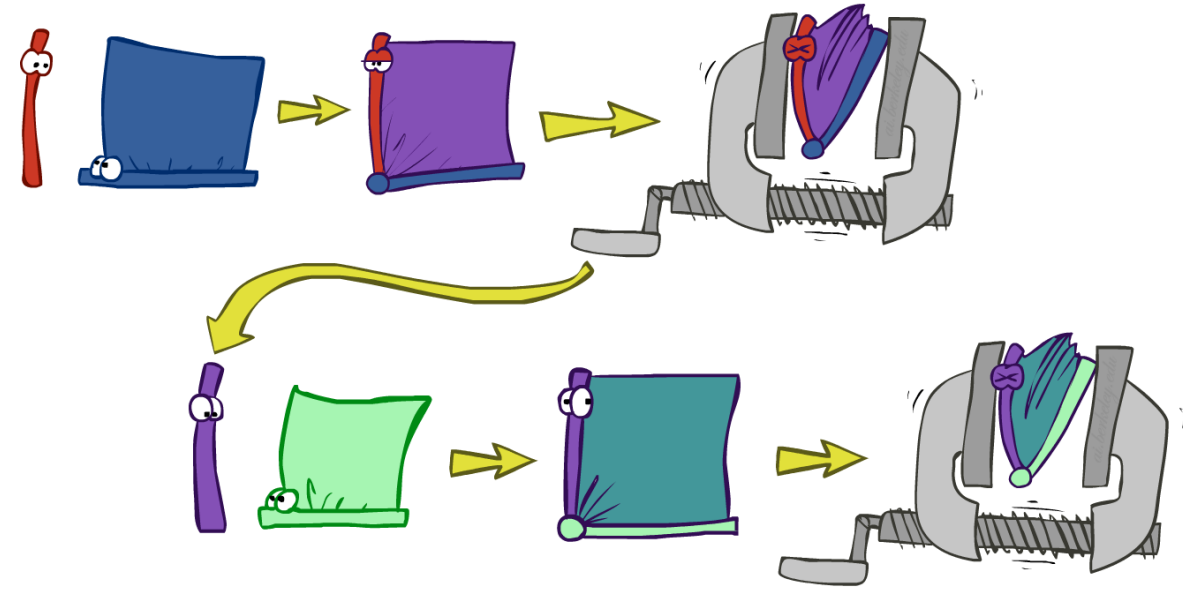
- Why is inference by enumeration so slow?

- You join up the whole joint distribution before you sum out the hidden variables

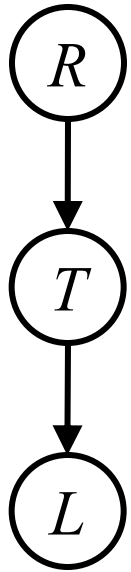


- Idea: interleave joining and marginalizing!

- Called “Variable Elimination”
- Still NP-hard, but usually much faster than inference by enumeration



Traffic Domain



$$P(L) = ?$$

■ Inference by Enumeration

$$= \sum_t \sum_r \underbrace{P(L|t)P(r)P(t|r)}_{\text{Join on } r}$$
$$\underbrace{\hspace{10em}}_{\text{Join on } t}$$
$$\underbrace{\hspace{10em}}_{\text{Eliminate } r}$$
$$\underbrace{\hspace{10em}}_{\text{Eliminate } t}$$

■ Variable Elimination

$$= \sum_t P(L|t) \underbrace{\sum_r P(r)P(t|r)}_{\text{Join on } r}$$
$$\underbrace{\hspace{10em}}_{\text{Eliminate } r}$$
$$\underbrace{\hspace{10em}}_{\text{Join on } t}$$
$$\underbrace{\hspace{10em}}_{\text{Eliminate } t}$$

VE: Computational and Space Complexity

- Does there always exist an ordering that only results in small factors?
 - No!
 - NP-Hard!

Bayes' Nets

- ✓ Representation
- ✓ Conditional Independences
- Probabilistic Inference
 - ✓ Enumeration (exact, exponential complexity)
 - ✓ Variable elimination (exact, worst-case exponential complexity, often better)
 - ✓ Inference is NP-complete
 - Sampling (approximate)

Approximate Inference: Sampling

- Sampling is a lot like repeated simulation

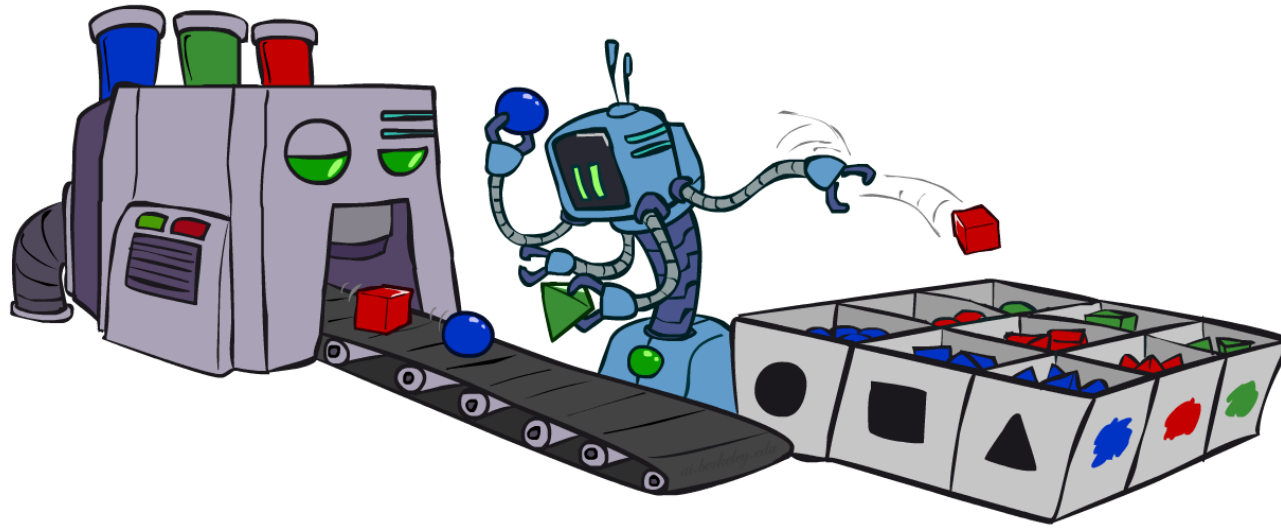
- Predicting the weather, basketball games, ...

- Basic idea

- Draw N samples from a sampling distribution S
 - Compute an approximate posterior probability
 - Show this converges to the true probability P

- Why sample?

- Learning: get samples from a distribution you don't know
 - Inference: getting a sample is faster than computing the right answer (e.g. with variable elimination)



Sampling

- Sampling from given distribution

- Step 1: Get sample u from uniform distribution over $[0, 1)$

```
>>> import random
>>> random.random()
0.6303136415860905
```

- Step 2: Convert this sample u into an outcome for the given distribution by having each outcome associated with a sub-interval of $[0,1)$ with sub-interval size equal to probability of the outcome

- Example

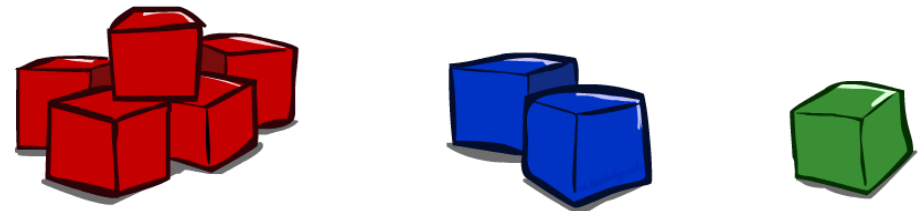
C	P(C)
red	0.6
green	0.1
blue	0.3

$$0 \leq u < 0.6, \rightarrow C = \text{red}$$

$$0.6 \leq u < 0.7, \rightarrow C = \text{green}$$

$$0.7 \leq u < 1, \rightarrow C = \text{blue}$$

- If `random()` returns $u = 0.83$, then our sample is $C = \text{blue}$
- E.g, after sampling 8 times:

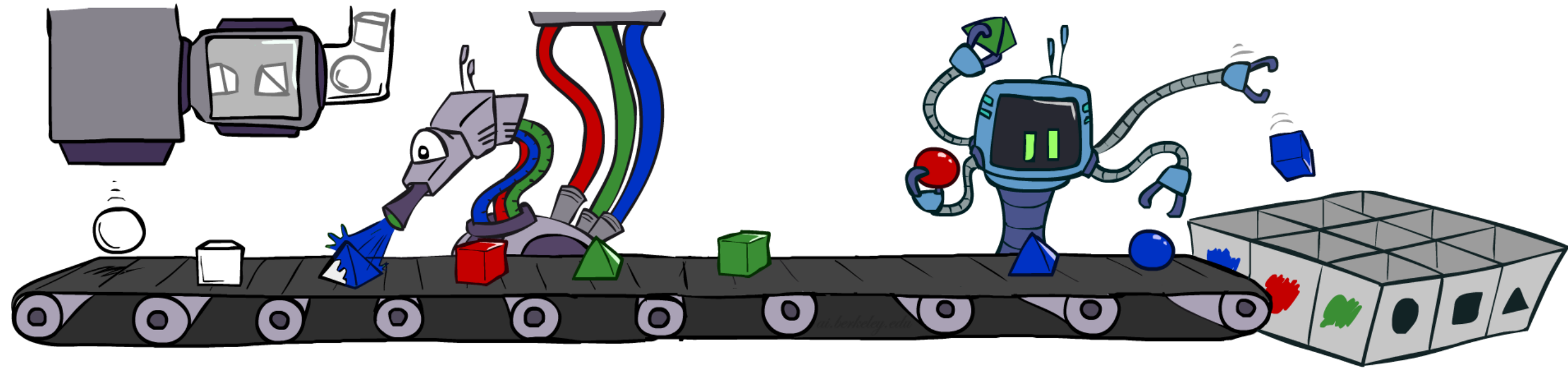


Sampling in Bayes' Nets

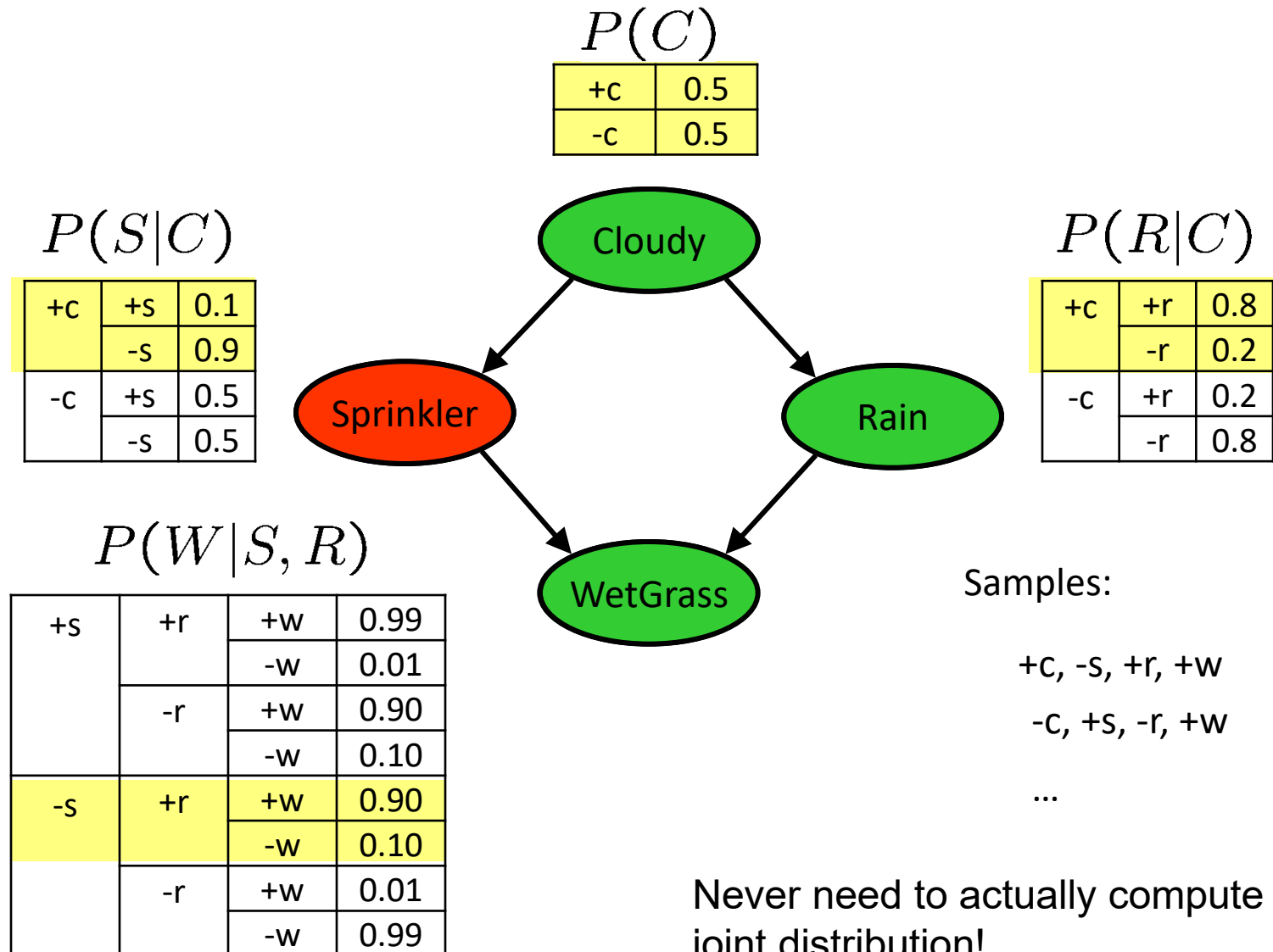
- Prior Sampling
- Rejection Sampling
- Likelihood Weighting
- Gibbs Sampling

Prior Sampling

- Sample a bunch of samples using the conditional probability tables.
- Then use these samples to compute any desired probabilistic query.



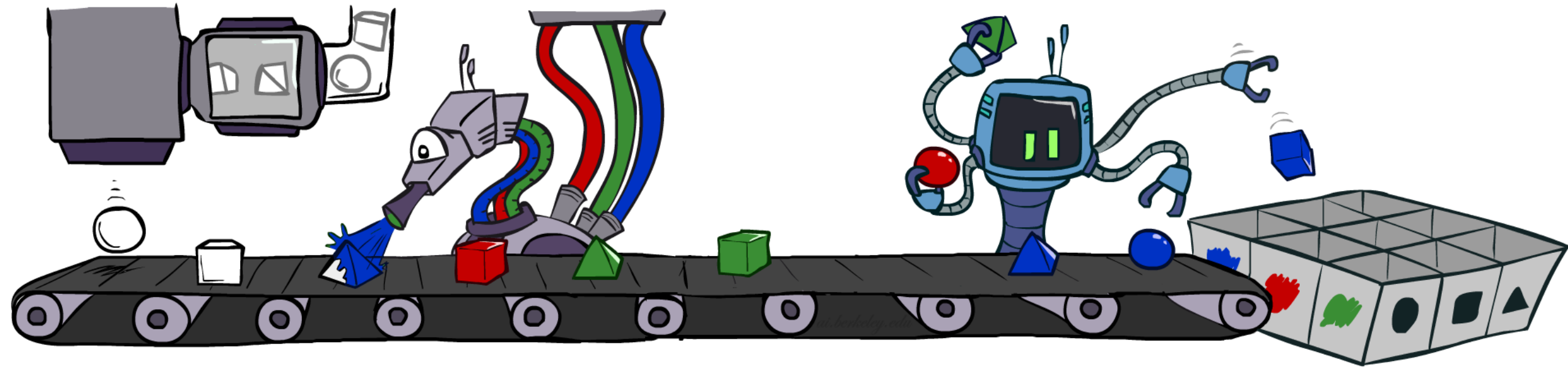
Prior Sampling



Never need to actually compute the full joint distribution!

Prior Sampling

- For $i=1, 2, \dots, n$
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- Return $(x_1, x_2, \dots, x_n)$



Prior Sampling

- This process generates samples with probability:

$$S_{PS}(x_1 \dots x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(X_i)) = P(x_1 \dots x_n)$$

...i.e. the BN's joint probability

- Let the number of samples of an event be $N_{PS}(x_1 \dots x_n)$
- Then
$$\begin{aligned} \lim_{N \rightarrow \infty} \hat{P}(x_1, \dots, x_n) &= \lim_{N \rightarrow \infty} N_{PS}(x_1, \dots, x_n) / N \\ &= S_{PS}(x_1, \dots, x_n) \\ &= P(x_1 \dots x_n) \end{aligned}$$
- I.e., the sampling procedure is **consistent**

Example

- We'll get a bunch of samples from the BN:

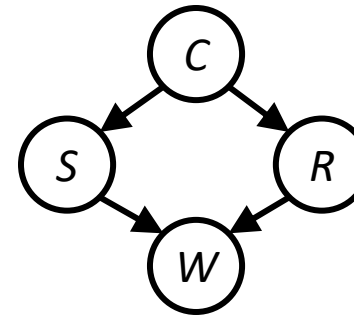
+c, -s, +r, +w

+c, +s, +r, +w

-c, +s, +r, -w

+c, -s, +r, +w

-c, -s, -r, +w



- What is $P(W)$?

- We have counts $\langle +w:4, -w:1 \rangle$
- Normalize to get $P(W) = \langle +w:0.8, -w:0.2 \rangle$
- This will get closer to the true distribution with more samples
- Can estimate anything else, too
- Fast: can use fewer samples if less time (what's the drawback?)

Practice

- We'll get a bunch of samples from the BN:

+c, -s, +r, +w

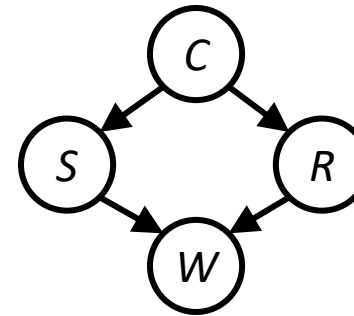
+c, +s, +r, +w

-c, +s, +r, -w

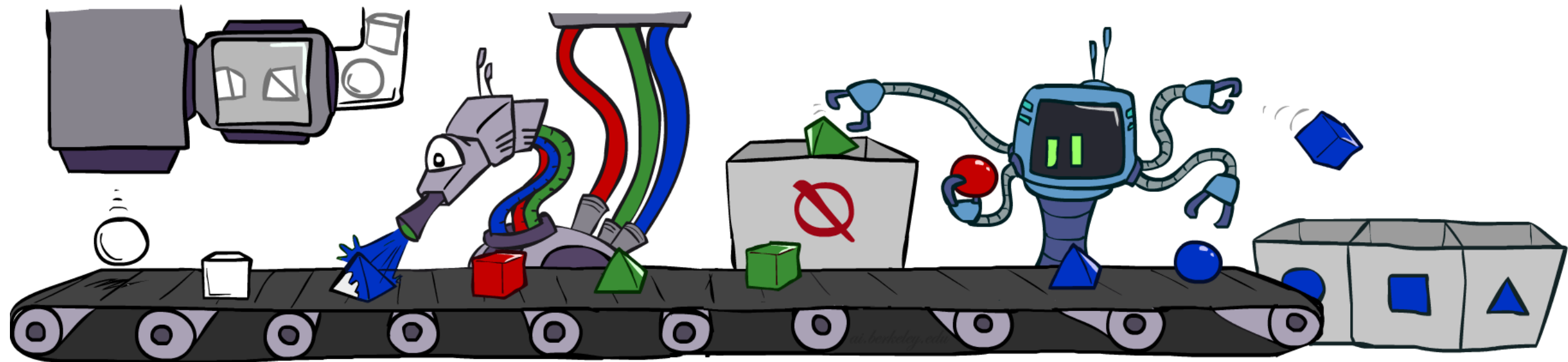
+c, -s, +r, +w

-c, -s, -r, +w

- $P(C \mid +w)$?
- $P(C \mid +r, +w)$?
- $P(C \mid -r, -w)$?

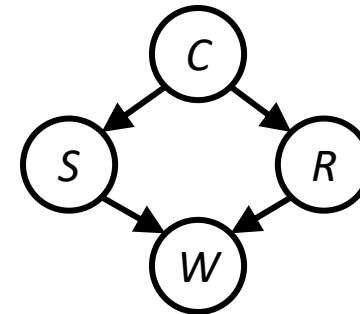


Rejection Sampling



Rejection Sampling

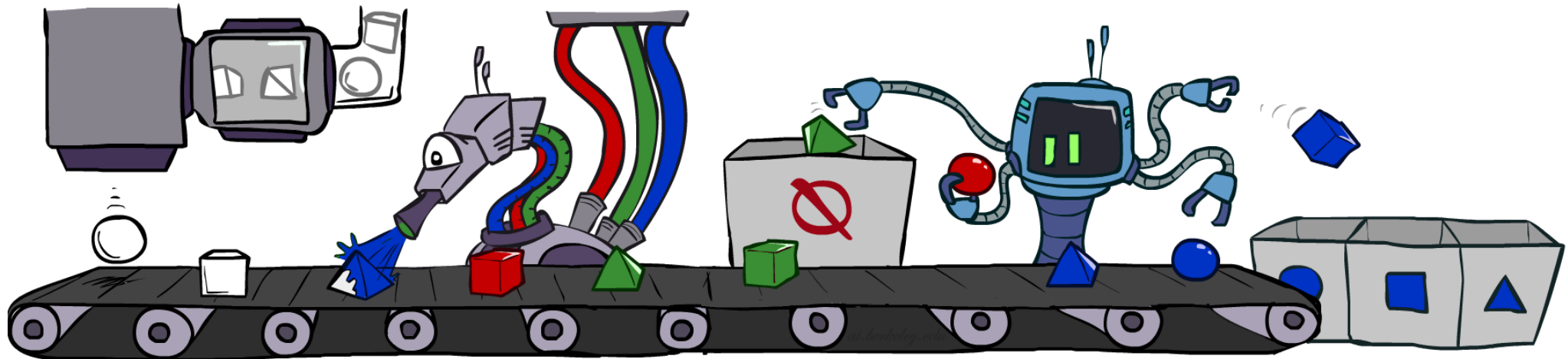
- Let's say we want $P(C)$
 - No point keeping all samples around
 - Just tally counts of C as we go
- Let's say we want $P(C | +s)$
 - Same thing: tally C outcomes, but ignore (reject) samples which don't have $S=+s$
 - This is called rejection sampling
 - It is also consistent for conditional probabilities (i.e., correct in the limit)



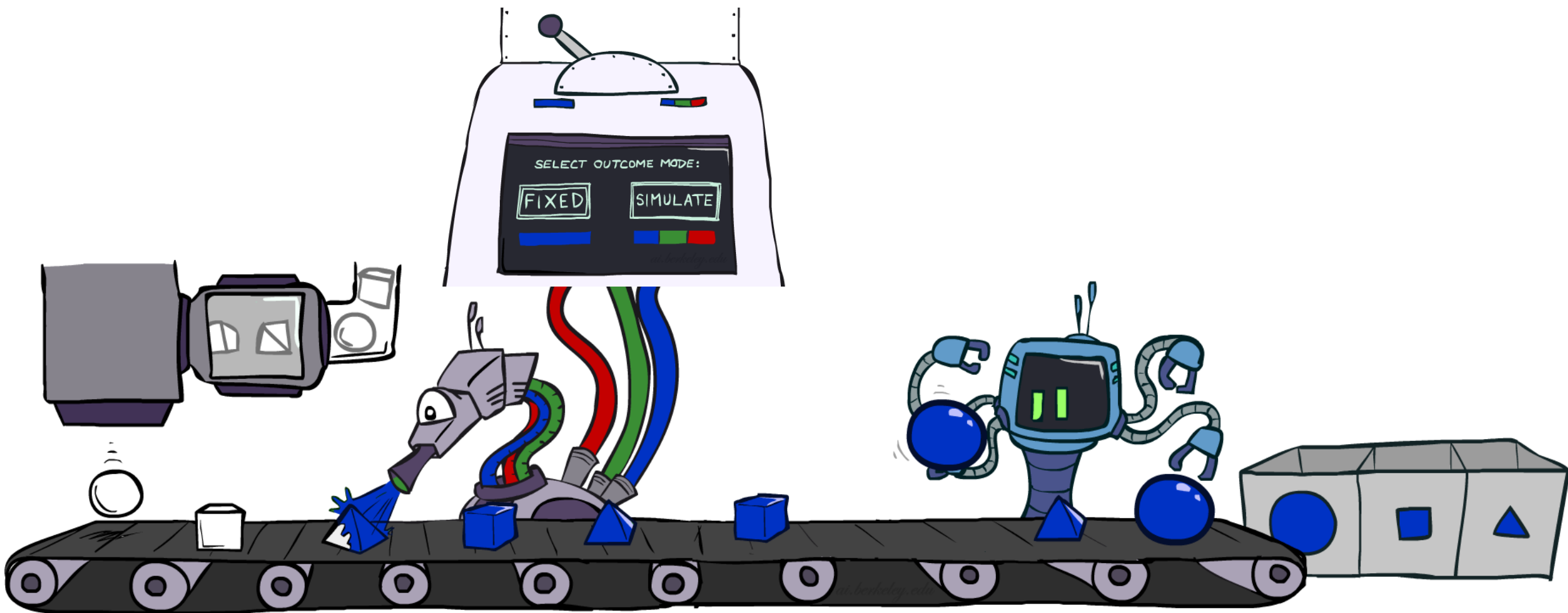
+c, -s, +r, +w
+c, +s, +r, +w
-c, +s, +r, -w
+c, -s, +r, +w
-c, -s, -r, +w

Rejection Sampling

- IN: evidence instantiation
- For $i=1, 2, \dots, n$
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
 - If x_i not consistent with evidence
 - Reject: Return, and no sample is generated in this cycle
- Return $(x_1, x_2, \dots, x_n)$

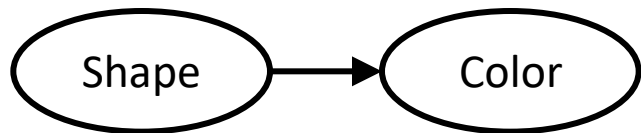


Likelihood Weighting

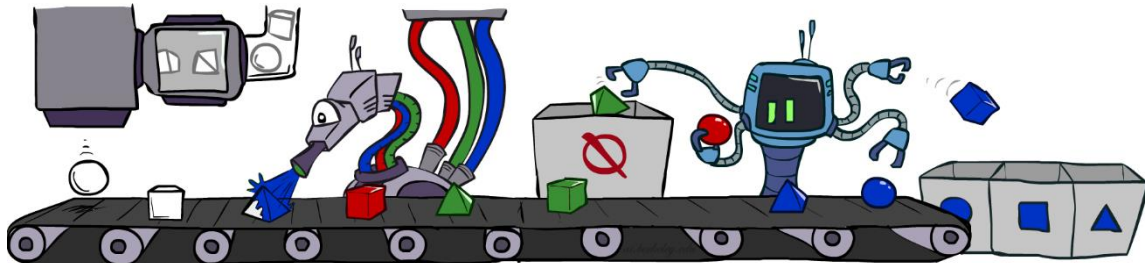


Likelihood Weighting

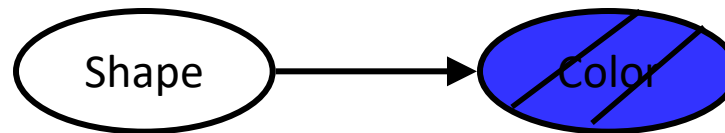
- Problem with rejection sampling:
 - If evidence is unlikely, rejects lots of samples
 - Evidence not exploited as you sample
 - Consider $P(\text{Shape} | \text{blue})$



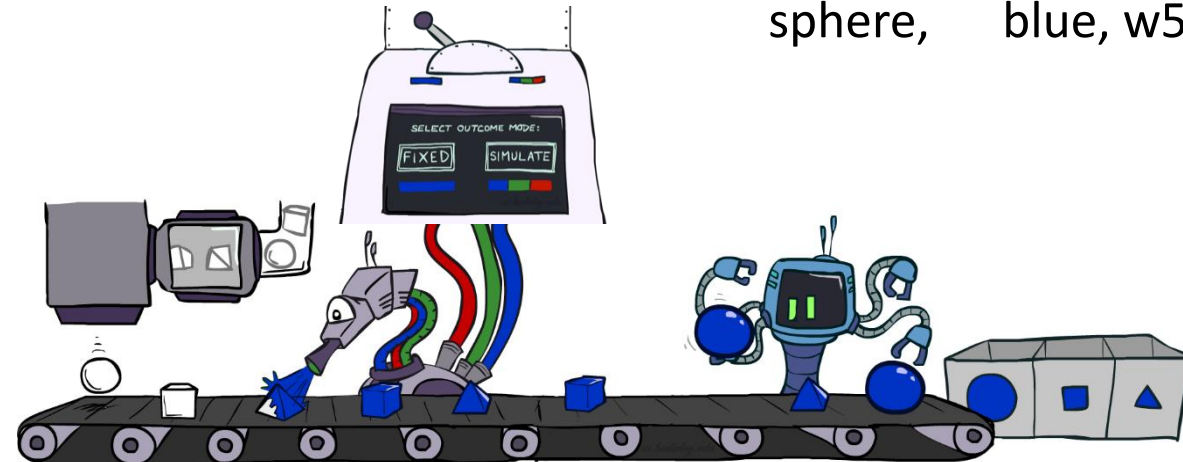
~~pyramid, green~~
~~pyramid, red~~
sphere, blue
cube, red
~~sphere, green~~



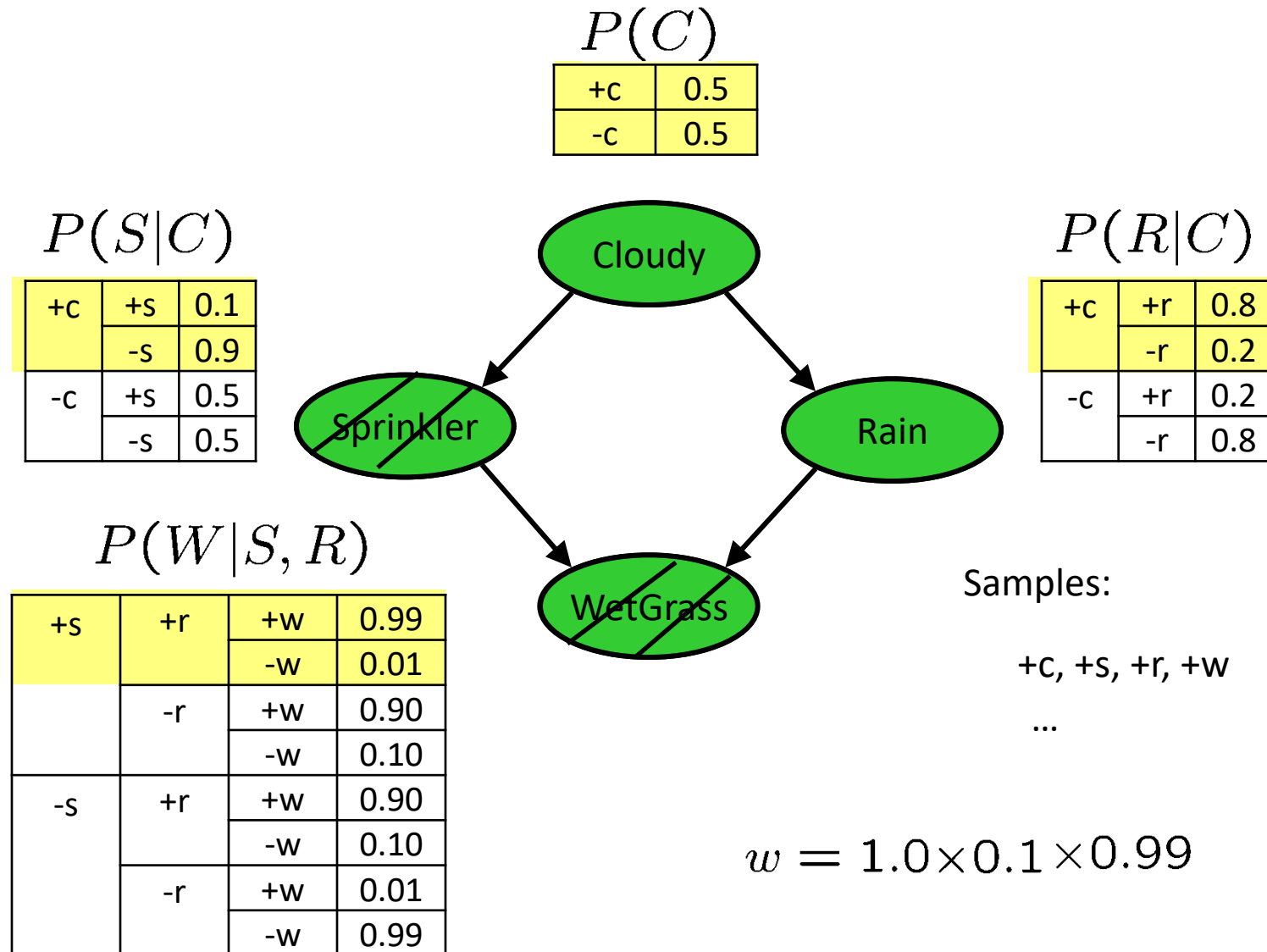
- Idea: fix evidence variables and sample the rest
 - Problem: sample distribution not consistent!
 - Solution: weight by probability of evidence given parents



pyramid, blue, w1
pyramid, blue, w2
sphere, blue, w3
cube, blue, w4
sphere, blue, w5



Likelihood Weighting

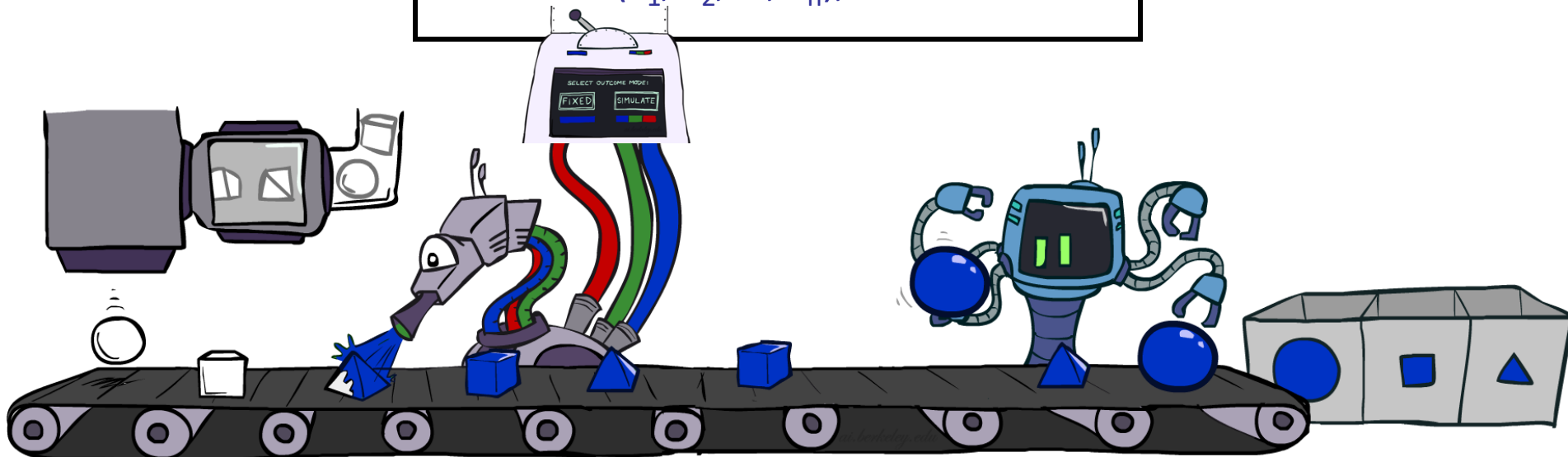


Likelihood Weighting

- IN: evidence instantiation
- $w = 1.0$
- for $i=1, 2, \dots, n$
 - if X_i is an evidence variable
 - $X_i = \text{observation } x_i \text{ for } X_i$
 - Set $w = w * P(x_i \mid \text{Parents}(X_i))$
 - else
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- return $(x_1, x_2, \dots, x_n), w$

Now each sample doesn't count as 1.0 but has a weight. Need to take a weighted average.

$P(Q|\text{Evidence}) =$
Sum(weights of samples consistent with Query) / Total Weight of All samples.



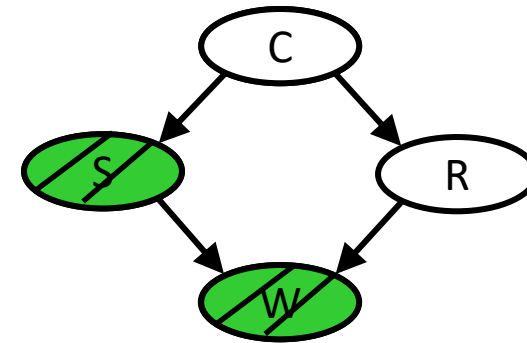
Likelihood Weighting

- Sampling distribution if z sampled and e fixed evidence

$$S_{WS}(z, e) = \prod_{i=1}^l P(z_i | \text{Parents}(Z_i))$$

- Now, samples have weights

$$w(z, e) = \prod_{i=1}^m P(e_i | \text{Parents}(E_i))$$

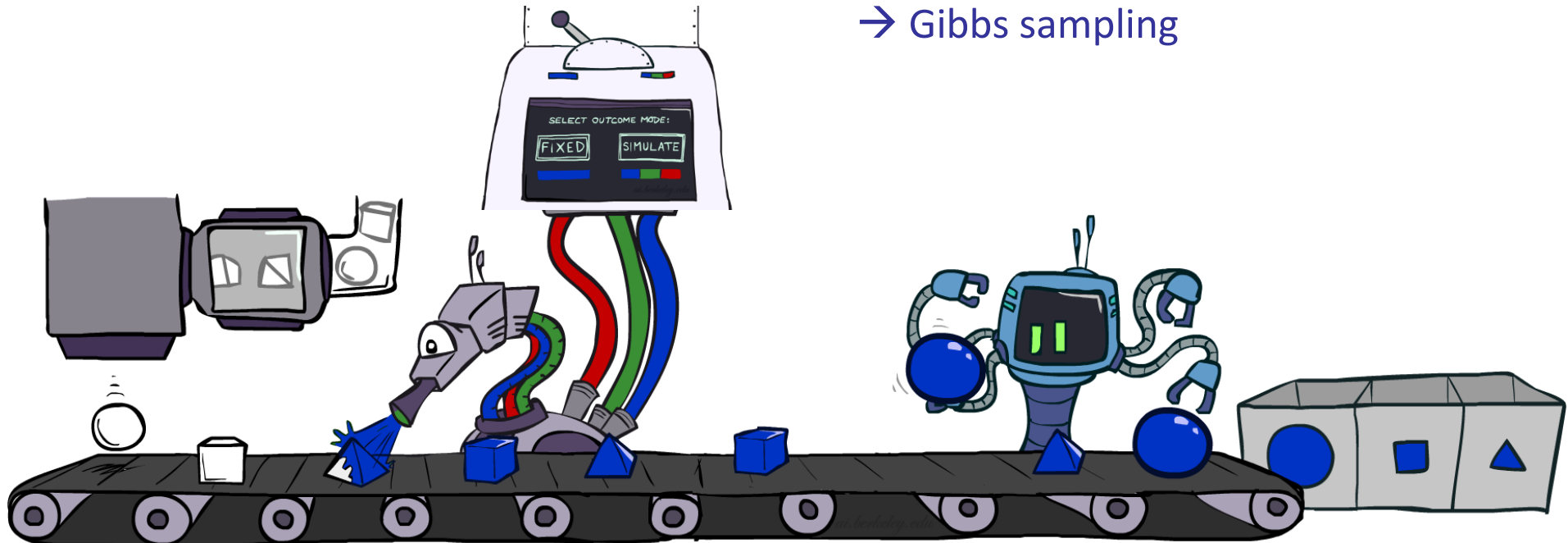


- Together, weighted sampling distribution is consistent

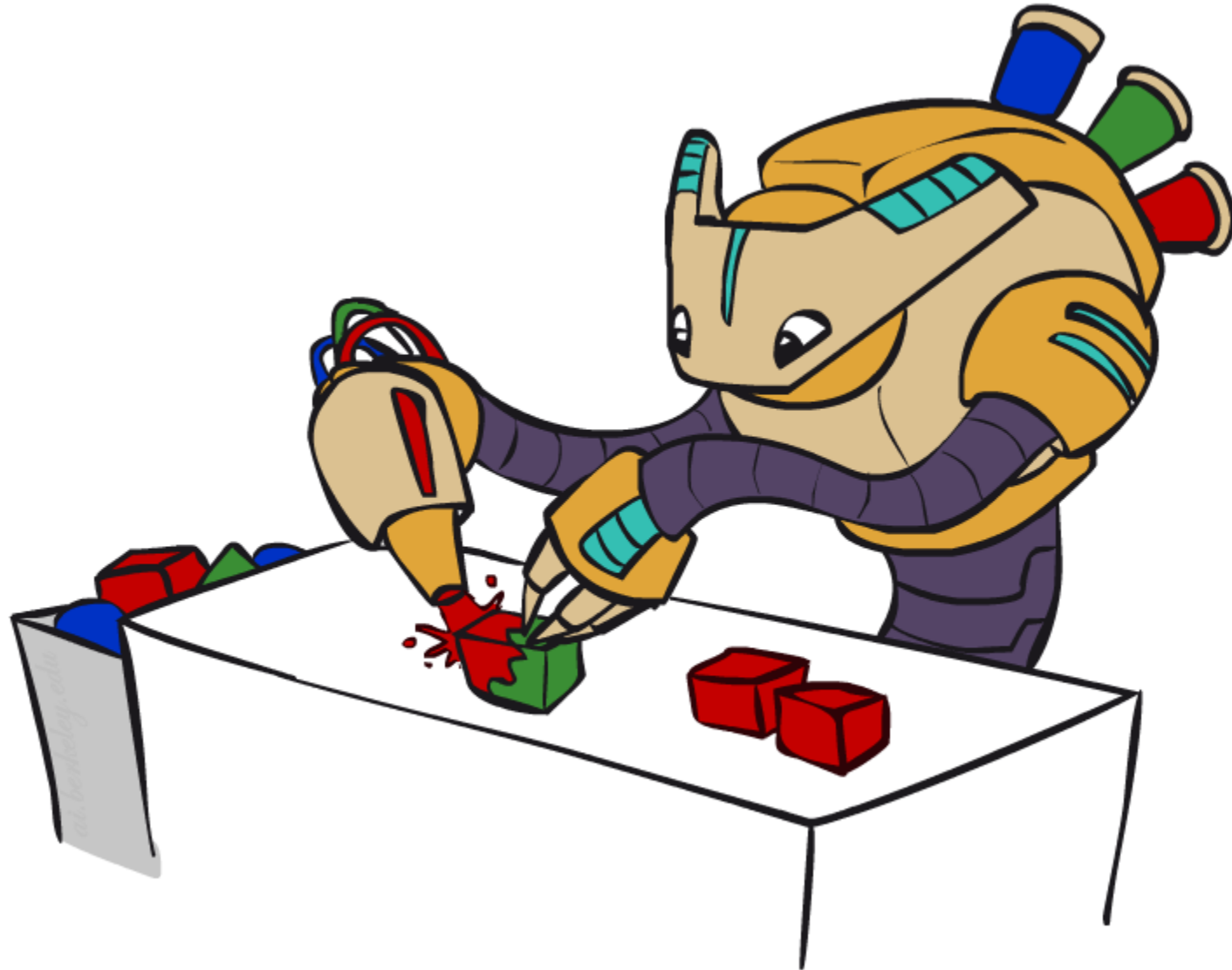
$$\begin{aligned} S_{WS}(z, e) \cdot w(z, e) &= \prod_{i=1}^l P(z_i | \text{Parents}(z_i)) \prod_{i=1}^m P(e_i | \text{Parents}(e_i)) \\ &= P(z, e) \end{aligned}$$

Likelihood Weighting

- Likelihood weighting is good
 - We have taken evidence into account as we generate the sample
 - E.g. here, W 's value will get picked based on the evidence values of S , R
 - More of our samples will reflect the state of the world suggested by the evidence
- Likelihood weighting doesn't solve all our problems (why?)
 - Evidence influences the choice of downstream variables, but not upstream ones (C isn't more likely to get a value matching the evidence)
- We would like to consider evidence when we sample every variable
 - Gibbs sampling



Gibbs Sampling



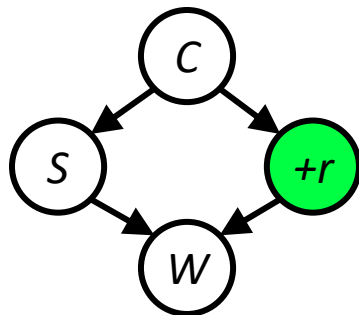
Gibbs Sampling

- *Procedure:* keep track of a full instantiation $x_1, x_2, \dots, x_n$. Start with an arbitrary instantiation consistent with the evidence. Sample one variable at a time, conditioned on all the rest, but keep evidence fixed. Keep repeating this for a long time (infinite in theory).
- *Property:* in the limit of repeating this infinitely many times the resulting sample is coming from the correct distribution. No need to weight!
- *Rationale:* both upstream and downstream variables condition on evidence.
- In contrast: likelihood weighting only conditions on upstream evidence, and hence weights obtained in likelihood weighting can sometimes be very small. Sum of weights over all samples is indicative of how many “effective” samples were obtained, so want high weight.

Gibbs Sampling Example: $P(S \mid +r)$

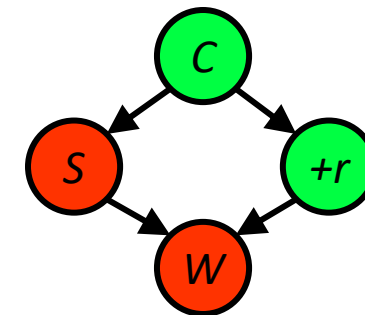
- Step 1: Fix evidence

- $R = +r$



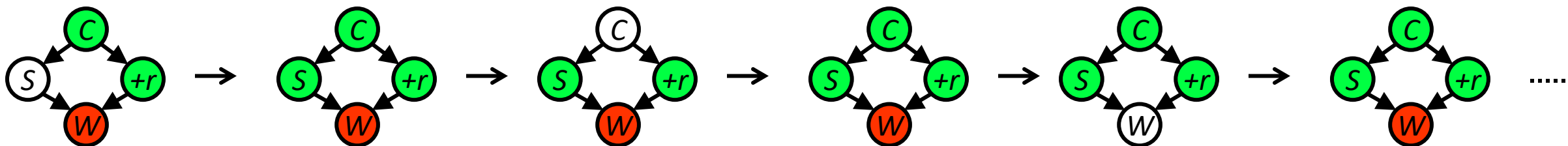
- Step 2: Initialize other variables

- Randomly



- Steps 3: Repeat

- Randomly choose a non-evidence variable X
 - Resample X from $P(X \mid \text{all other variables})$



Sample from $P(S \mid +c, -w, +r)$

Sample from $P(C \mid +s, -w, +r)$

Sample from $P(W \mid +s, +c, +r)$

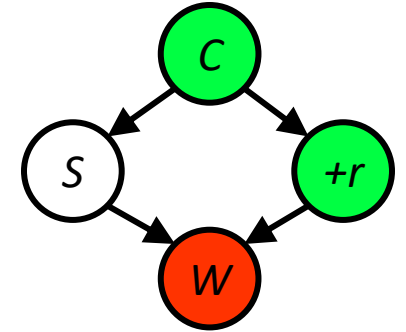
Gibbs Sampling

- How is this better than sampling from the full joint?
 - In a Bayes' Net, sampling a variable given all the other variables (e.g. $P(R|S,C,W)$) is usually much easier than sampling from the full joint distribution
 - Only requires a join on the variable to be sampled (in this case, a join on R)
 - The resulting factor only depends on the variable's parents, its children, and its children's parents (this is often referred to as its Markov blanket)

Efficient Resampling of One Variable

- Sample from $P(S \mid +c, +r, -w)$

$$\begin{aligned} P(S \mid +c, +r, -w) &= \frac{P(S, +c, +r, -w)}{P(+c, +r, -w)} \\ &= \frac{P(S, +c, +r, -w)}{\sum_s P(s, +c, +r, -w)} \\ &= \frac{P(+c)P(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{\sum_s P(+c)P(s \mid +c)P(+r \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(+c)P(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{P(+c)P(+r \mid +c) \sum_s P(s \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(S \mid +c)P(-w \mid S, +r)}{\sum_s P(s \mid +c)P(-w \mid s, +r)} \end{aligned}$$



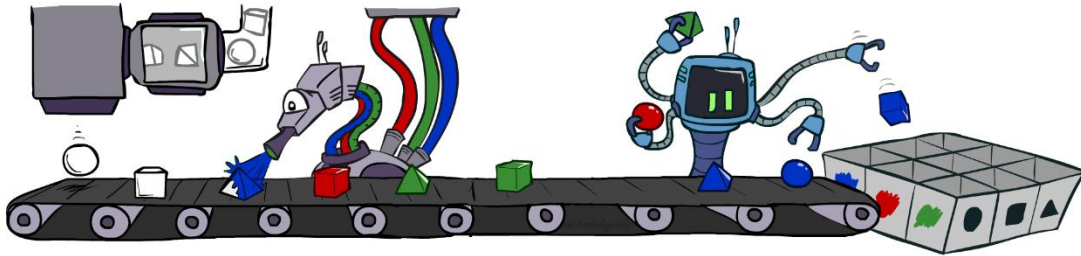
- Many things cancel out – only CPTs with S remain!
- More generally: only CPTs that have resampled variable need to be considered

Further Notes on Gibbs Sampling

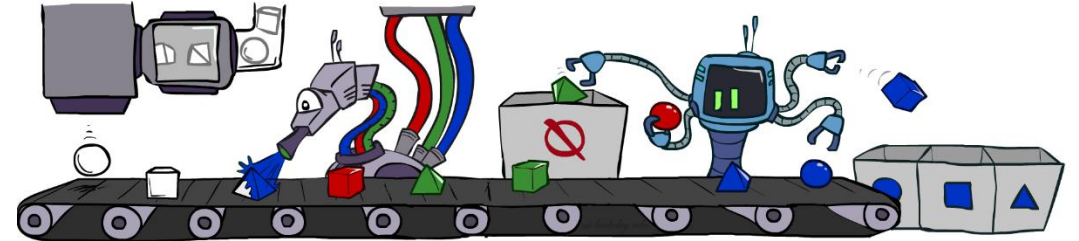
- Gibbs sampling produces sample from the query distribution $P(Q | e)$ in limit of re-sampling infinitely often
- Gibbs sampling is a special case of more general methods called Markov chain Monte Carlo (MCMC) methods
 - Metropolis-Hastings is one of the more famous MCMC methods (in fact, Gibbs sampling is a special case of Metropolis-Hastings)
- You may read about Monte Carlo methods – they're just sampling

Bayes' Net Sampling Summary

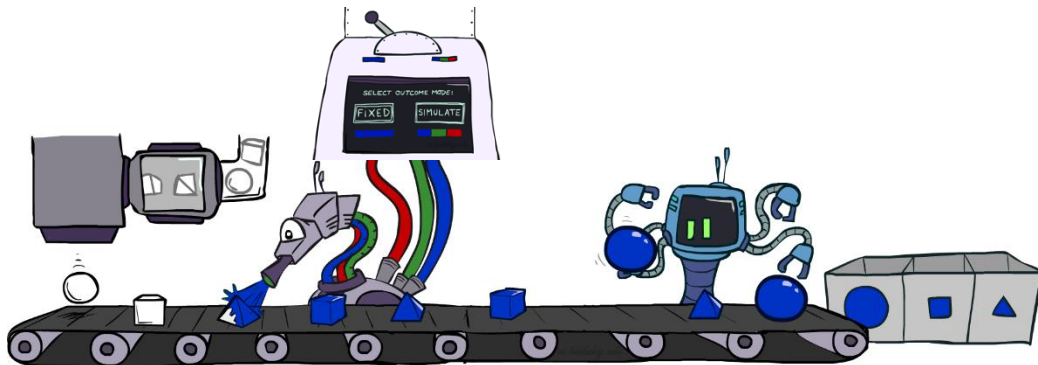
- Prior Sampling P



- Rejection Sampling $P(Q | e)$



- Likelihood Weighting $P(Q | e)$



- Gibbs Sampling $P(Q | e)$

